VTPEH 6270 Final Report

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1 Introduction, Research Question, Plausibility and Context

1.1 Introduction

This analysis uses the COVID-19 Vaccine Hesitancy for County and Local Estimates data set published by the U.S. Centers for Disease Control and Prevention (CDC) [1]. The dataset contains county-level estimates of vaccine hesitancy among adults, vaccination coverage as of June 10, 2021, and measures of social vulnerability, including the Social Vulnerability Index (SVI) and demographic characteristics. Each observation represents a U.S. county and includes both continuous and categorical variables suitable for exploratory data analysis and visualization. The complete data set and accompanying information are available at https://data.cdc.gov/Vaccinations/Vaccine-Hesitancy-for-COVID-19-County-and-local-es/q9mh-h2tw/about_data

1.2 Research Question:

> **Is COVID-19 vaccine hesitancy higher in counties with higher Social Vulnerability Index (SVI)?**

- **Variable A (Outcome, continuous):** Estimated hesitant — percentage of adults estimated to be hesitant toward COVID-19 vaccination
- **Variable B (Predictor, categorical):** SVI category — categorized measure of county-level social vulnerability

1.3 Scientific Plausibility and Public Health Context:

COVID-19 vaccine hesitancy directly undermined efforts to reduce transmission and mortality during the pandemic, making it one of the most consequential public health challenges of the past decade. The CDC/ATSDR Social Vulnerability Index (SVI) captures community-level conditions — socioeconomic status, minority status, household composition, and housing/transportation access — that shape how communities experience and respond to public health crises [2].

There are strong reasons to expect higher SVI to correlate with greater hesitancy: reduced healthcare access, structural barriers to vaccination, and historically rooted mistrust of medical institutions among communities that have faced documented mistreatment [3, 4]. Sociodemographic and socioeconomic characteristics, along with health literacy, have been shown to be important predictors of vaccine hesitancy [5]. If this association holds, public health agencies can target outreach and resources to high-SVI counties to close vaccination gaps and reduce disparities in future pandemic responses.

2 Material & Methods

2.1 Loading Data and Data Description

2.1.1 Loading Data

```r
library(readxl)
library(dplyr)
library(ggplot2)
library(knitr)
library(kableExtra)
library(tidyr)
```

```r
# Set working directory
setwd("/Users/sowmya427/Desktop/VTPEH-6270-R-Lab/data")

# Load file
raw <- read.csv("R Lab_Vaccine_Hesitancy.csv", check.names = FALSE)

# Confirm columns loaded correctly
cat("Column names:\n")
print(names(raw))

df <- raw %>%
  rename(
    fips               = `FIPS Code`,
    county             = `County Name`,
    state              = `State`,
    hesitant           = `Estimated hesitant`,
    hesitant_or_unsure = `Estimated hesitant or unsure`,
    strongly_hesitant  = `Estimated strongly hesitant`,
    svi                = `Social Vulnerability Index (SVI)`,
    svi_cat            = `SVI Category`,
    vax_rate           = `Percent adults fully vaccinated against COVID-19 (as of 6/10/21)`,
    pct_hispanic       = `Percent Hispanic`,
    pct_black          = `Percent non-Hispanic Black`,
```

```r
    pct_white          = `Percent non-Hispanic White`
  ) %>%
  mutate(
    hesitant_pct = as.numeric(gsub("%", "", hesitant)),
    vax_rate_pct = as.numeric(gsub("%", "", vax_rate)),
    svi_cat = factor(
      svi_cat,
      levels = c(
        "Very Low Vulnerability",
        "Low Vulnerability",
        "Moderate Vulnerability",
        "High Vulnerability",
        "Very High Vulnerability"
      )
    )
  ) %>%
  filter(!is.na(svi), !is.na(hesitant_pct))

cat("Final sample size:", nrow(df), "counties\n")
```

2.1.2 Nature Theme

```r
dir.create("figures/CP06", recursive = TRUE, showWarnings = FALSE)

theme_nature <- function(base_size = 10) {
  theme_classic(base_size = base_size) +
    theme(
      text             = element_text(family = "sans"),
      axis.title       = element_text(size = base_size, color = "black"),
      axis.text        = element_text(size = base_size - 2, color = "black"),
      axis.line        = element_line(color = "black", linewidth = 0.5),
      axis.ticks       = element_line(color = "black", linewidth = 0.5),
      legend.text      = element_text(size = base_size - 2),
      legend.title     = element_text(size = base_size - 1, face = "bold"),
      legend.key.size  = unit(0.4, "cm"),
      plot.title       = element_text(size = base_size + 1, face = "bold", hjust = 0),
      plot.caption     = element_text(size = base_size - 3, color = "gray50", hjust = 0),
      panel.border     = element_blank(),
      strip.background = element_blank()
    )
}
```

2.1.3 Description Table

```r
summary_tbl <- df %>%
  group_by(svi_cat) %>%
  summarise(
    n      = n(),
    Mean   = round(mean(hesitant_pct), 2),
    SD     = round(sd(hesitant_pct),   2),
```

```r
    Median = round(median(hesitant_pct), 2),
    Min    = round(min(hesitant_pct),  2),
    Max    = round(max(hesitant_pct),  2),
    .groups = "drop"
  )

kable(
  summary_tbl,
  col.names = c("SVI Category", "n", "Mean (\\%)", "SD",
                "Median (\\%)", "Min (\\%)", "Max (\\%)"),
  caption  = "Descriptive statistics of estimated COVID-19 vaccine hesitancy (\\%) by SVI category.",
  booktabs = TRUE
) %>%
  kable_styling(
    latex_options = c("hold_position", "striped"),
    font_size     = 10
  )
```

Table 1: Descriptive statistics of estimated COVID-19 vaccine hesitancy (%) by SVI category.

SVI Category	n	Mean (\%)	SD	Median (\%)	Min (\%)	Max (\%)
Very Low Vulnerability	628	11.53	4.62	10.66	2.69	26.7
Low Vulnerability	628	12.28	4.43	12.35	3.75	26.7
Moderate Vulnerability	628	13.11	4.42	12.99	4.10	26.7
High Vulnerability	628	14.13	4.33	13.83	4.42	26.7
Very High Vulnerability	629	15.26	4.42	15.56	5.20	26.7

2.2 Statistical Analysis

To examine whether COVID-19 vaccine hesitancy rates differ across Social Vulnerability Index (SVI) categories, the following statistical approach was used.

Assumption Checks

Prior to selecting a statistical test, the normality of the outcome variable (estimated vaccine hesitancy rate, %) was assessed using the Shapiro-Wilk test, where a p-value < 0.05 indicates a significant departure from normality. A histogram and Q-Q plot were also used to visually inspect the distribution of the data. Normality is a required assumption for parametric tests such as one-way ANOVA, and failure to meet this assumption requires the use of non-parametric alternatives.

Model Selection

Because the Shapiro-Wilk test confirmed that the hesitancy rate data were non-normally distributed, a Kruskal-Wallis rank-sum test was selected to compare hesitancy rates across the five SVI categories (Very Low, Low, Moderate, High, and Very High Vulnerability). The Kruskal-Wallis test is the non-parametric equivalent of a one-way ANOVA and does not require the assumption of normality. In addition, Spearman's rank correlation was used to quantify the strength and direction of the monotonic association between the continuous SVI score and vaccine hesitancy rate. Spearman's correlation was chosen over Pearson's correlation as it does not assume normality and is appropriate for non-normally distributed continuous variables.

Post-Hoc Analysis

A significant Kruskal-Wallis result indicates that at least one SVI category differs from the others, but does not specify which pairs. Therefore, pairwise Wilcoxon rank-sum tests were conducted as a post-hoc analysis to identify which specific SVI category pairs showed statistically significant differences in vaccine hesitancy rates. P-values were adjusted using the Bonferroni correction to control for multiple comparisons and reduce the risk of Type I error. Statistical significance was set at alpha = 0.05 for all analyses.

2.3 Code

All code and data used in this analysis are available at: https://github.com/SS0908/VTPEH-6270-R-Lab

2.4 Interactive Shiny Application

An interactive Shiny dashboard accompanying this report is available at: https://ss0908.shinyapps.io/ShinyAppHW1/

The application allows users to explore the relationship between Social Vulnerability Index (SVI) and COVID-19 vaccine hesitancy interactively, including filtering by SVI category and viewing county-level distributions.

3 Results

3.1 Data Visualization

3.1.1 Boxplot

```r
p_box <- ggplot(df, aes(x = svi_cat, y = hesitant_pct, fill = svi_cat)) +
  geom_boxplot(
    alpha         = 0.80,
    outlier.size  = 0.6,
    outlier.alpha = 0.35,
    linewidth     = 0.4
  ) +
  scale_fill_manual(
    name   = "SVI Category",
    values = c(
      "Very Low Vulnerability"  = "#4393C3",
      "Low Vulnerability"       = "#92C5DE",
      "Moderate Vulnerability"  = "#D9D9D9",
      "High Vulnerability"      = "#F4A582",
      "Very High Vulnerability" = "#D6604D"
    )
  ) +
  scale_x_discrete(
    labels = c("Very Low", "Low", "Moderate", "High", "Very High")
  ) +
  labs(
    title   = "COVID-19 Vaccine Hesitancy by Social Vulnerability Category",
    x       = "SVI Category",
    y       = "Estimated Hesitancy (%)",
    caption = "Source: CDC Vaccine Hesitancy County Estimates"
```

```
  ) +
  theme_nature()

p_box
```

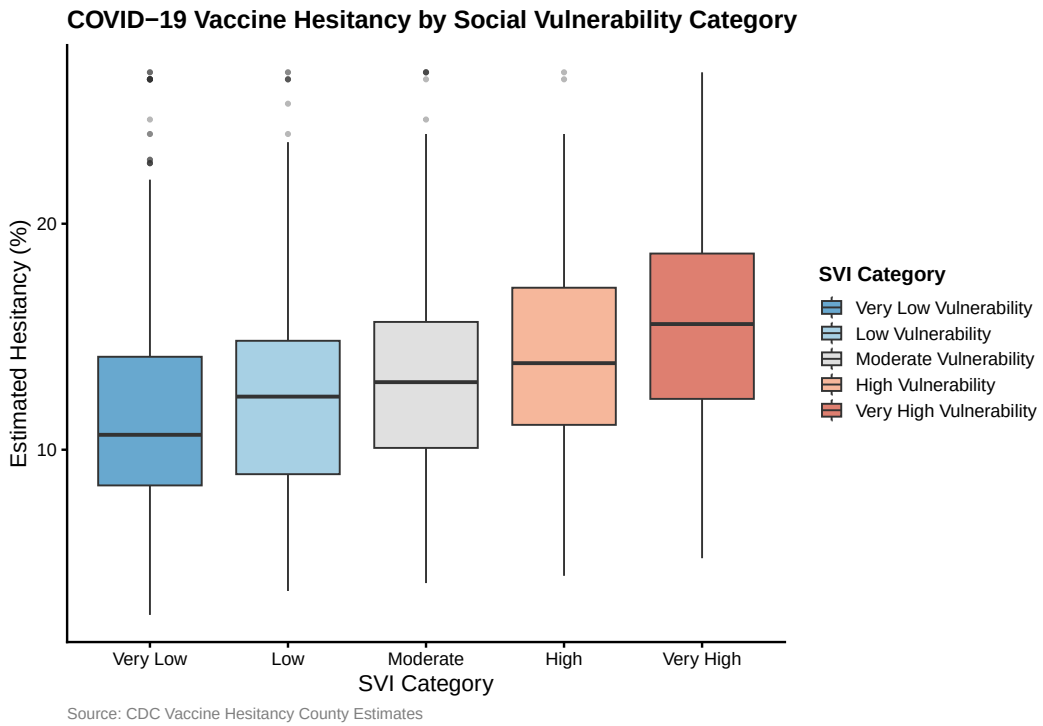

Figure 1: Distribution of estimated COVID-19 vaccine hesitancy (%) by SVI category. Boxes span the interquartile range; horizontal lines show medians; dots are outliers.

3.1.2 Scatterplot

```
p_scatter <- ggplot(df, aes(x = svi, y = hesitant_pct)) +
  geom_point(
    aes(color = "County data"),
    alpha = 0.15,
    size  = 0.6
  ) +
  geom_smooth(
    aes(color = "LOESS smoother"),
    method    = "loess",
    linewidth = 1,
    se        = TRUE,
    fill      = "#F4A582",
    alpha     = 0.3
  ) +
  scale_color_manual(
    name   = "Legend",
    values = c(
```

```
      "County data"    = "#4393C3",
      "LOESS smoother" = "#D6604D"
    )
  ) +
  labs(
    title   = "Social Vulnerability Index vs. COVID-19 Vaccine Hesitancy",
    x       = "Social Vulnerability Index (SVI, 0-1)",
    y       = "Estimated Hesitancy (%)",
    caption = "Source: CDC Vaccine Hesitancy County Estimates"
  ) +
  theme_nature()

p_scatter
```

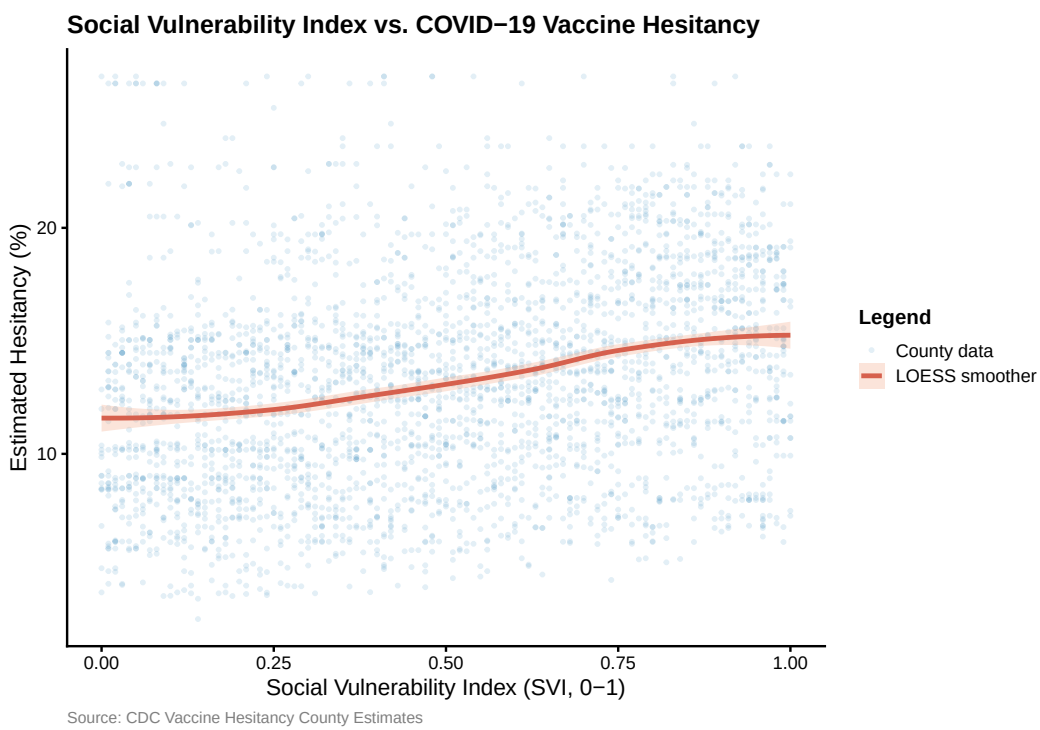

Figure 2: Association between continuous SVI score and estimated vaccine hesitancy (%) across U.S. counties (n = 3,141). Each point is one county. Red curve is a LOESS smoother with 95% CI shaded.

3.1.3 Normality; Plot 1 - Histogram

```
p_hist <- ggplot(df, aes(x = hesitant_pct)) +
  geom_histogram(
    aes(fill = "Counties (n = 3,141)"),
    bins  = 40,
    color = "white",
    alpha = 0.85
  ) +
  geom_vline(
```

```r
    aes(xintercept = mean(hesitant_pct, na.rm = TRUE), color = "Mean"),
    linewidth = 1,
    linetype  = "dashed"
  ) +
  geom_vline(
    aes(xintercept = median(hesitant_pct, na.rm = TRUE), color = "Median"),
    linewidth = 1,
    linetype  = "dotted"
  ) +
  scale_fill_manual(
    name   = "Distribution",
    values = c("Counties (n = 3,141)" = "#4393C3")
  ) +
  scale_color_manual(
    name   = "Reference Lines",
    values = c("Mean" = "#D6604D", "Median" = "#4DAF4A")
  ) +
  labs(
    title = "Distribution of COVID-19 Vaccine Hesitancy Rates",
    x     = "Estimated Hesitancy (%)",
    y     = "Number of Counties"
  ) +
  theme_nature() +
  theme(legend.position = "right")

p_hist
```

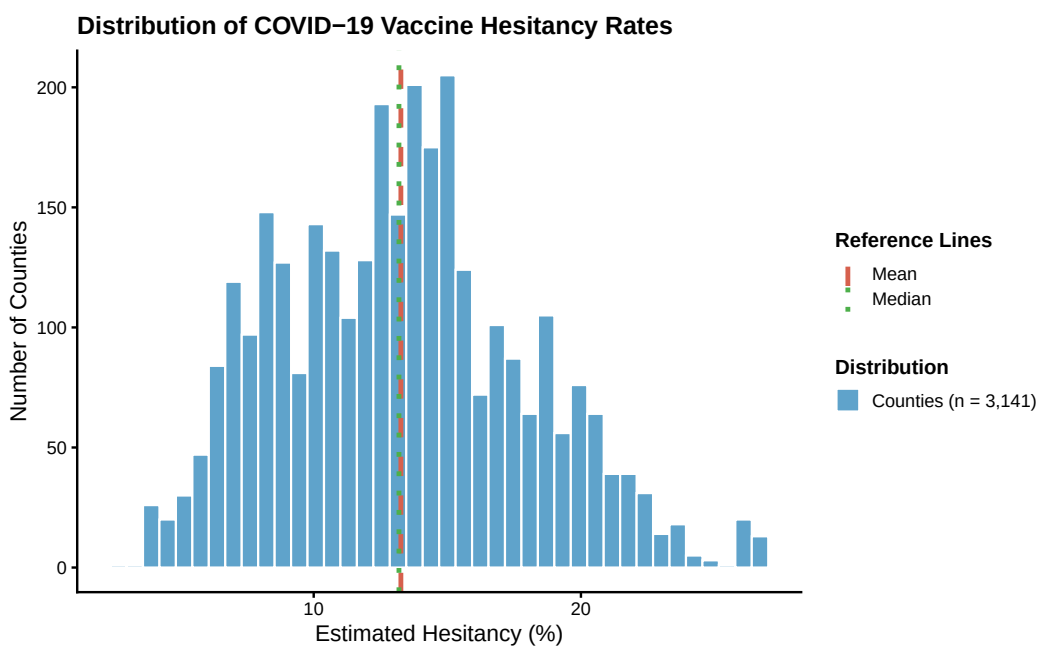

Figure 3: Histogram of estimated COVID-19 vaccine hesitancy rates (%) across U.S. counties. Dashed red line = mean; dotted green line = median.

3.1.4 Normality; Plot 2 - Q-Q plot

```r
p_qq <- ggplot(df, aes(sample = hesitant_pct)) +
  stat_qq(
    aes(color = "Sample data"),
    alpha = 0.3,
    size  = 0.8
  ) +
  stat_qq_line(
    aes(color = "Normal reference line"),
    linewidth = 1
  ) +
  scale_color_manual(
    name   = "Legend",
    values = c(
      "Sample data"           = "#4393C3",
      "Normal reference line" = "#D6604D"
    )
  ) +
  labs(
    title = "Q-Q Plot of Vaccine Hesitancy Rates",
    x     = "Theoretical Quantiles",
    y     = "Sample Quantiles (%)"
  ) +
  theme_nature() +
  theme(legend.position = "bottom")

p_qq
```

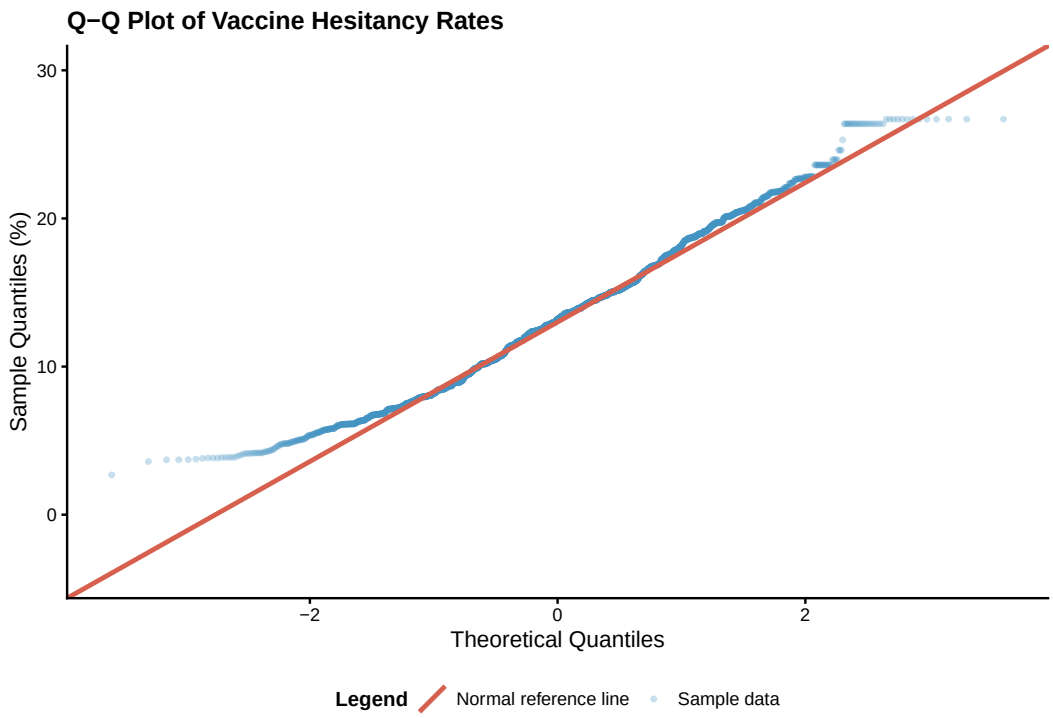

Figure 4: Q-Q plot of estimated vaccine hesitancy rates. Blue points = sample data; red line = expected values under a normal distribution. Deviations at the tails indicate non-normality.

3.2 Statistical Analysis

3.2.1 Shapiro-Wilk Test

```
sw <- shapiro.test(df$hesitant_pct)

sw_tbl <- data.frame(
  Test        = "Shapiro-Wilk",
  W           = round(sw$statistic, 4),
  `p-value`   = ifelse(sw$p.value < 0.001, "< 0.001", round(sw$p.value, 4)),
  Result      = ifelse(sw$p.value < 0.05,
                       "Non-normal - use non-parametric tests",
                       "Normal distribution"),
  check.names = FALSE
)

kable(
  sw_tbl,
  caption  = "Shapiro-Wilk normality test for vaccine hesitancy rates.",
  booktabs = TRUE,
  align    = c("l", "c", "c", "l")
) %>%
  kable_styling(
    latex_options = c("hold_position", "striped"),
```

```
    font_size     = 10
  )
```

Table 2: Shapiro-Wilk normality test for vaccine hesitancy rates.

	Test	W	p-value	Result
W	Shapiro-Wilk	0.9869	< 0.001	Non-normal - use non-parametric tests

The Shapiro-Wilk test returned a W statistic of 0.9869 and a p-value < 0.001, indicating that the distribution of vaccine hesitancy rates departs significantly from normality. This means we cannot assume the data follow a normal distribution, and therefore non-parametric tests must be used for all subsequent group comparisons rather than parametric alternatives such as ANOVA or a t-test.

3.2.2 Kruskal-Walis Test

```
kw <- kruskal.test(hesitant_pct ~ svi_cat, data = df)

kw_tbl <- data.frame(
  Test          = "Kruskal-Wallis Rank-Sum Test",
  `Chi-squared` = round(kw$statistic, 3),
  df            = kw$parameter,
  `p-value`     = ifelse(kw$p.value < 0.001, "< 0.001", round(kw$p.value, 4)),
  Result        = "Significant difference in hesitancy across SVI categories",
  check.names   = FALSE
)
rownames(kw_tbl) <- NULL

kable(
  kw_tbl,
  caption  = "Kruskal-Wallis test comparing vaccine hesitancy rates across SVI categories.",
  booktabs = TRUE,
  align    = c("l", "c", "c", "c", "l")
) %>%
  kable_styling(
    latex_options = c("hold_position", "striped", "scale_down"),
    font_size     = 10
  ) %>%
  column_spec(5, width = "5cm")
```

Table 3: Kruskal-Wallis test comparing vaccine hesitancy rates across SVI categories.

Test	Chi-squared	df	p-value	Result
Kruskal-Wallis Rank-Sum Test	284.461	4	< 0.001	Significant difference in hesitancy across SVI categories

The Kruskal-Wallis rank-sum test revealed a statistically significant difference in vaccine hesitancy rates across the five SVI categories (X^2 = 284.46, df = 4, p < 0.001). This result allows us to reject the null

hypothesis that hesitancy rates are equal across SVI categories. In other words, social vulnerability level is associated with differences in COVID-19 vaccine hesitancy at the county level. However, this test alone does not tell us which specific categories differ from one another, which is addressed by the pairwise comparisons below.

3.2.3 Pairwise Wilcoxon

```
pw <- pairwise.wilcox.test(
  x               = df$hesitant_pct,
  g               = df$svi_cat,
  p.adjust.method = "bonferroni"
)

# Format the p-value matrix into a clean table
pw_tbl <- as.data.frame(pw$p.value) %>%
  tibble::rownames_to_column(var = "Group") %>%
  mutate(across(where(is.numeric), ~ case_when(
    . < 0.001 ~ "< 0.001",
    . < 0.05  ~ as.character(round(., 3)),
    is.na(.)  ~ "--",
    TRUE      ~ as.character(round(., 3))
  )))

kable(
  pw_tbl,
  col.names = c("Group", "Very Low", "Low", "Moderate", "High"),
  caption  = "Pairwise Wilcoxon rank-sum tests comparing vaccine hesitancy across SVI categories (Bonfe
  booktabs = TRUE,
  align    = "c"
) %>%
  kable_styling(
    latex_options = c("hold_position", "scale_down", "striped"),
    font_size     = 9
  ) %>%
  column_spec(1, width = "4cm")
```

Table 4: Pairwise Wilcoxon rank-sum tests comparing vaccine hesitancy across SVI categories (Bonferroni correction). '–' = comparison not applicable.

Group	Very Low	Low	Moderate	High
Low Vulnerability	0.002	–	–	–
Moderate Vulnerability	< 0.001	0.006	–	–
High Vulnerability	< 0.001	< 0.001	< 0.001	–
Very High Vulnerability	< 0.001	< 0.001	< 0.001	< 0.001

Pairwise Wilcoxon rank-sum tests with Bonferroni correction were conducted to identify which specific SVI category pairs drive the overall significant Kruskal-Wallis result. The results show that most pairwise comparisons between SVI categories were statistically significant ($p < 0.001$) after correction for multiple testing. Counties in higher vulnerability categories consistently showed greater vaccine hesitancy than those in lower vulnerability categories. Comparisons between adjacent categories (e.g., Very Low vs. Low Vulnerability)

may show weaker or non-significant differences, reflecting the gradual nature of the relationship between social vulnerability and hesitancy.

3.2.4 Spearman Correlation

```
sp <- cor.test(df$svi, df$hesitant_pct, method = "spearman")

sp_tbl <- data.frame(
  Test        = "Spearman Rank Correlation",
  `rho (p)`   = round(sp$estimate, 3),
  S           = round(sp$statistic, 1),
  `p-value`   = ifelse(sp$p.value < 0.001, "< 0.001", round(sp$p.value, 4)),
  Direction   = "Positive association",
  Result      = "Higher SVI linked to higher hesitancy",
  check.names = FALSE
)
rownames(sp_tbl) <- NULL

kable(
  sp_tbl,
  caption  = "Spearman rank correlation between continuous SVI score and vaccine hesitancy (\\%).",
  booktabs = TRUE,
  align    = c("l", "c", "c", "c", "l", "l")
) %>%
  kable_styling(
    latex_options = c("hold_position", "scale_down", "striped"),
    font_size     = 10
  ) %>%
  column_spec(1, width = "3.5cm") %>%
  column_spec(5, width = "2.5cm") %>%
  column_spec(6, width = "3cm")
```

Table 5: Spearman rank correlation between continuous SVI score and vaccine hesitancy (%).

Test	rho (p)	S	p-value	Direction	Result
Spearman Rank Correlation	0.3	3615074674	< 0.001	Positive association	Higher SVI linked to higher hesitancy

Spearman's rank correlation between the continuous SVI score and vaccine hesitancy rate yielded a correlation coefficient of $p = 0.3$ ($p < 0.001$), indicating a statistically significant positive monotonic association. This means that as the Social Vulnerability Index increases across counties, estimated vaccine hesitancy also tends to increase. The magnitude of the correlation suggests a moderate positive relationship, supporting the conclusion that social vulnerability is meaningfully — though not perfectly — linked to COVID-19 vaccine hesitancy at the county level. This finding directly addresses our research question and is consistent with the patterns observed visually in the scatterplot and boxplot.

4 Conclusion

The results of this analysis support the hypothesis that COVID-19 vaccine hesitancy is higher in counties with greater social vulnerability. The Kruskal-Wallis test identified statistically significant differences in hesitancy rates across SVI categories (X^2 = 284.46, $p < 0.001$), and pairwise Wilcoxon comparisons confirmed that higher-vulnerability counties showed greater hesitancy than lower-vulnerability counties across most category pairs. The Spearman correlation ($p = 0.3$, $p < 0.001$) further supports a moderate positive association between the continuous SVI score and estimated hesitancy rate.

These findings align with the research question posed in the Introduction — counties with higher social vulnerability do tend to have higher estimated COVID-19 vaccine hesitancy. The moderate strength of the association suggests that while social vulnerability is a meaningful predictor of hesitancy, other factors likely contribute as well. These results highlight the importance of targeting socially vulnerable communities in public health vaccination campaigns.

5 AI Use Disclosure Statement

This document was created with the assistance of Claude (Anthropic) to help structure and generate the R code, which was reviewed and adjusted by the author. Interpretations and written sections were composed by the author.

6 References

1. Centers for Disease Control and Prevention. Vaccine Hesitancy for COVID-19: County and Local Estimates. https://data.cdc.gov/Vaccinations/Vaccine-Hesitancy-for-COVID-19-County-and-local-es/q9mh-h2tw/about_data

2. Centers for Disease Control and Prevention/Agency for Toxic Substances and Disease Registry (CDC/ATSDR). CDC/ATSDR Social Vulnerability Index (SVI). Atlanta, GA: U.S. Department of Health and Human Services. https://www.atsdr.cdc.gov/place-health/php/svi/index.html

3. Lee J, Huang Y. COVID-19 Vaccine Hesitancy: The Role of Socioeconomic Factors and Spatial Effects. *Vaccines*. 2022;10(3):352. doi:10.3390/vaccines10030352

4. Nguyen VT, et al. Factors related to COVID-19 vaccine hesitancy among middle- and low-income adults in the U.S. *J Epidemiol Community Health*. 2023;77(5):328-335.

5. Lamot M, Kirbiš A. Understanding Vaccine Hesitancy: A Comparison of Sociodemographic and Socioeconomic Predictors with Health Literacy Dimensions. *Vaccines*. 2024;12(10):1141. doi:10.3390/vaccines12101141